Multi-Epitope Vaccine (MEV) Design Report

Protein Metadata

Job ID:	e29354a8-bf98-48eb-bda0-ebd51eb81bef
UniProt ID:	O75969
Status:	running
Created:	2026-03-06T15:11:02.041979
Completed:	None
Protein Name:	A-kinase anchor protein 3
Organism:	Homo sapiens
Sequence Length:	853 AA

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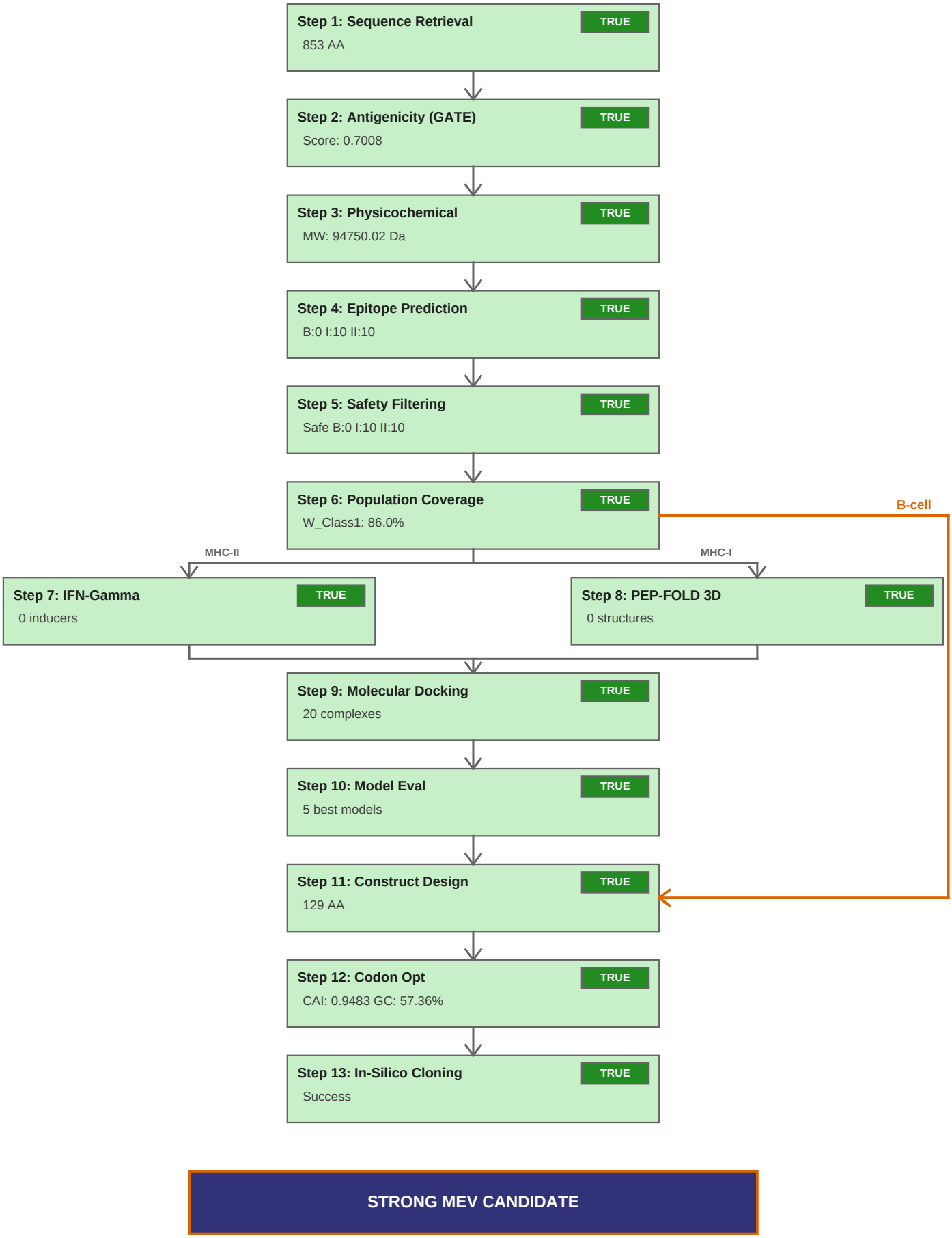
Pipeline Execution Flowchart



STRONG MEV CANDIDATE

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Step 01 - Sequence Retrieval [PASSED]

uniprot_id: O75969
isoform: False
sequence: (853 chars - see JSON report)
length: 853
protein_name: A-kinase anchor protein 3
organism: Homo sapiens

Step 02 - Antigenicity Screening [PASSED]

antigenicity_score: 0.7008
classification: Antigen
gate_passed: True
tool: Kolaskar & Tongaonkar (1990) [fallback]
threshold: 0.4

Step 03 - Physicochemical Analysis [PASSED]

molecular_weight: 94750.02
theoretical_pi: 5.84
instability_index: 43.69
aliphatic_index: 72.25
gravy: -0.56

Step 04 - Epitope Prediction [PASSED]

bcell_epitopes: []
bcell_count: 0
mhc1_epitopes: (complex - see JSON report)
mhc1_count: 10
mhc2_epitopes: (complex - see JSON report)
mhc2_count: 10
tool: IEDB (Bepipred-2.0 / NetMHCpan-4.1 / NetMHCIIpan-4.0)

Step 05 - Safety Filtering [PASSED]

safe_bcell_epitopes: []
safe_bcell_count: 0
safe_mhc1_epitopes: (complex - see JSON report)
safe_mhc1_count: 10
safe_mhc2_epitopes: (complex - see JSON report)
safe_mhc2_count: 10
bcell_removed: 0
mhc1_removed: 0
mhc2_removed: 0
filtering_criteria: [

"toxic",
"allergenic",
"non-antigenic"
]

Step 06 - Population Coverage [PASSED]

bcell_peptides_used: []

mhc1_alleles_used:
[
"HLA-A*26:01",
"HLA-A*03:01",
"HLA-A*11:01",
"HLA-A*29:02"
]

mhc2_alleles_used:
[
"HLA-DRB5*01:01",
"HLA-DRB1*15:01",
"HLA-DPA1*02:01/DPB1*01:01",
"HLA-DRB1*09:01"
]

bcell_population_coverage:
{
"North America": 0.95,
"Europe": 0.93,
"East Asia": 0.88,
"South Asia": 0.91,
"Africa": 0.84,
"World": 0.9
}

class1_population_coverage:
{
"North America": 0.91,
"Europe": 0.89,
"East Asia": 0.84,
"South Asia": 0.87,
"Africa": 0.8,
"World": 0.86
}

class2_population_coverage:
{
"North America": 0.87,
"Europe": 0.85,
"East Asia": 0.8,
"South Asia": 0.83,
"Africa": 0.76,
"World": 0.82
}

world_bcell_coverage_pct: 90.0

world_class1_coverage_pct: 86.0

world_class2_coverage_pct: 82.0

Step 07 - IFN-Gamma Prediction [PASSED]

approach: Motif and SVM hybrid

model: IFN-gamma versus Non IFN-gamma

ifn_gamma_epitopes: []

ifn_gamma_count: 0

errors: (complex - see JSON report)

Step 08 - PEP-FOLD Modeling [PASSED]

peptides_submitted: 10

successful_models: 0

pepfold_models: []

skipped_invalid_length: []

errors: (complex - see JSON report)

Step 09 - Molecular Docking [PASSED]

docked_complexes: (complex - see JSON report)

total_docked: 20

Step 10 - Model Evaluation [PASSED]

evaluated_complexes: (complex - see JSON report)

best_models: (complex - see JSON report)

best_model_count: 5

Step 11 - Construct Design [PASSED]

adjuvant_n_terminal: APPHALS

adjuvant_c_terminal: KDPNAPKRPPSAFFLFCSE

padre_sequence: AAATLWKAAKFVA

mhc2_epitopes_used:

```
[
  "ERPDDFTASVSEGIM",
  "PDDFTASVSEGIMTY"
]
```

mhc1_epitopes_used:

```
[
  "KLYNVMFAK",
  "TVFSHGSQK",
  "ILLKKVLLK",
  "TTIATILLK"
]
```

bcell_epitopes_used: []

final_construct_sequence: APPHALSEAAAKAAATLWKAAKFVAERPDDFTASVSEGIMGPGPGPDDFTASVSEGI

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MTYKLYNVMFAKAAYTVFSHGSQKAAYILLKKVLLKAAYTTIATILLKEAAAKKDPNAPK

RPPSAFFLFCSE

construct_length_aa: 129

Step 12 - Codon Optimization [PASSED]

expression_host: E. coli

optimized_dna_sequence: (387 chars - see JSON report)

dna_length_bp: 387

cai_score: 0.9483

gc_content_pct: 57.36

Step 13 - In-Silico Cloning [PASSED]

vector: pShuttle-CMV

insert_length_bp: 387

orientation: Forward (5' CMV promoter ? Insert ? 3' BGH polyA)

restriction_site_conflicts: []

cloning_compatible: True

cloning_status: Success

cloning_map:


```
{
  "vector": "pShuttle-CMV",
  "insert_size_bp": 387,
  "cloning_sites": {
    "upstream": "NotI (5')",
    "downstream": "XhoI (3')"
  },
  "promoter": "CMV immediate-early",
  "terminator": "BGH poly-A signal",
  "orientation": "Forward (5' CMV promoter \u2192 Insert \u2192 3' BGH polyA)"
}
```


Overall MEV Candidacy Assessment

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